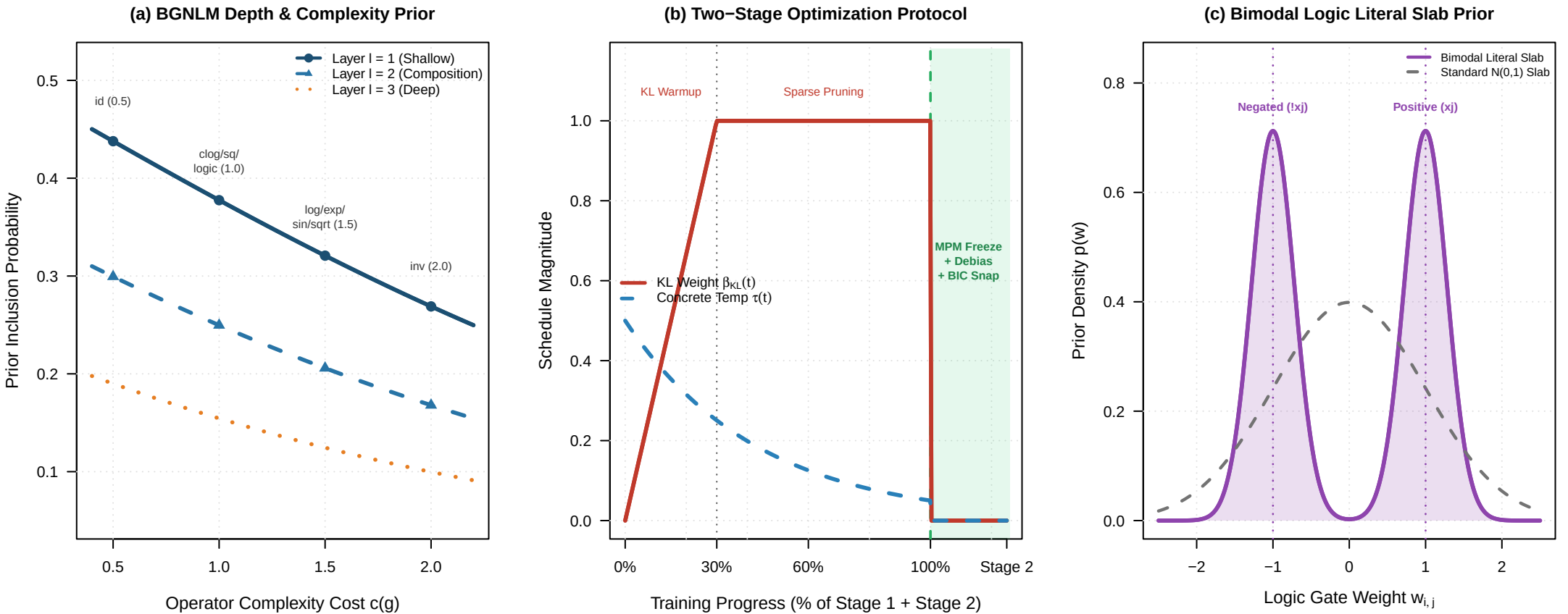
(a) BGNLM Depth & Complexity Prior
Layer l = 1 (Shallow)
Layer l = 2 (Composition)
Layer l = 3 (Deep)
id (0.5)
clog/sq/
logic (1.0)
log/exp/
sin/sqrt (1.5)
inv (2.0)
Prior Inclusion Probability
Operator Complexity Cost c(g)
(b) Two−Stage Optimization Protocol
KL Warmup
Sparse Pruning
MPM Freeze
+ Debias
+ BIC Snap
KL Weight βKL(t)
Concrete Temp τ(t)
Schedule Magnitude
0%
30%
60%
100%
Stage 2
Training Progress (% of Stage 1 + Stage 2)
(c) Bimodal Logic Literal Slab Prior
Bimodal Literal Slab
Standard N(0,1) Slab
Negated (!xj)
Positive (xj)
Prior Density p(w)
Logic Gate Weight wi, j